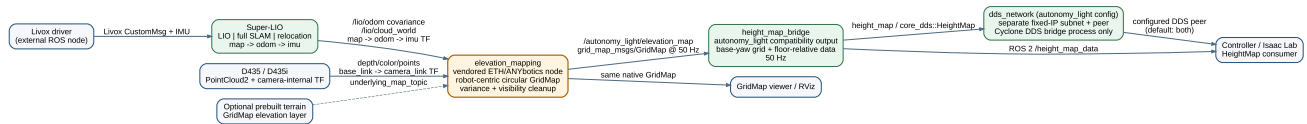


autonomy_light software architecture

Super-LIO + ETH/ANYbotics robot-centric elevation mapping · 2026-08-04

Design decision. ETH/ANYbotics owns elevation fusion. This repository owns calibrated bringup and one compact output bridge: native GridMap is always published at the configured rate, then converted at the same rate to the previous ROS 2 HeightMap or Cyclone DDS `core_dds::HeightMap` contract.



Node responsibilities

Node	Owner	Responsibility
Livox driver	external	Publishes one Livox CustomMsg and IMU stream.
Super-LIO	external	LiDAR–IMU odometry, ESKF covariance, full SLAM, saved-map relocation, and map → odom → imu.
D435/D435i driver	external	Publishes one PointCloud2 stream with its calibrated TF chain.
elevation_mapping	vendored ETH/ANYbotics	Robot-centric circular grid, sensor-specific variance, pose-covariance prediction, Bayesian fusion, and visibility cleanup.
height_map_bridge	this repository	50 Hz GridMap-to-legacy conversion, base-yaw resampling, local-floor-relative data, ROS 2/Cyclone DDS output selection.
autonomy_light	this repository	Configuration, process lifecycle, calibrated static TF, output transport and direct-DDS network contract.

Data and TF contract

Direction	Topic / frame	Meaning
input	/lio/odom	Super-LIO pose and 6×6 covariance for motion uncertainty propagation.
input	/lio/cloud_world	Registered LiDAR cloud, transformed to the map at its timestamp.
optional input	/camera/.../depth/color/points	D435/D435i cloud; directly fused with the structured-light model.

output	<code>/autonomy_light/elevation_map</code>	Native <code>grid_map_msgs/msg/GridMap</code> at configured 50 Hz, layers: elevation, upper/lower bound, uncertainty range.
output	<code>/autonomy_light/height_map_data</code>	Legacy <code>autonomy_light/msg/HeightMap</code> , base-yaw row-major array at the same rate.
output	<code>height_map</code>	Cyclone DDS <code>core_dds::HeightMap { sequence<float> data; };</code> default transport: both publishes this and ROS 2 together.
TF	<code>map → odom → imu → base_link → lidar_link</code> <code>base_link → camera_link → camera_depth_optical_frame</code>	Super-LIO owns dynamic transforms; launcher owns the calibrated static mounts, while the D435 driver owns camera-internal TF.

The local grid follows `base_link` in `map` with Grid Map's circular buffer. It preserves overlapping posterior cells while moving and clears only newly exposed strips. Unobserved terrain remains `NaN`; no arbitrary startup floor value is inserted. The compatibility bridge alone maps unknown cells to its legacy configured fill value (`0.48` by default).

Remote DDS network

`config/autonomy_light.yaml` owns `dds_network`: mode, NIC, fixed local IP, peer and multicast. With the default direct-DDS output, the launcher verifies the address and that it does not overlap `livox_network.subnet (192.168.1.0/24)` or its configured NIC, then adds `192.168.20.10/24` only to the configured computer-link NIC when missing. It passes a generated Cyclone DDS interface/peer configuration only to `height_map_bridge`; ROS 2, Livox and Super-LIO retain their own middleware/network behavior. The remote computer uses `192.168.20.1/24` and peer `192.168.20.10`. Explicit peers preserve discovery when Wi-Fi multicast is filtered.

Uncertainty and cleanup

Each input uses its own laser or D435 noise model. At each cloud timestamp the elevation node obtains the Super-LIO covariance, predicts map uncertainty due to robot motion, and applies the measurement update in the map frame. Its visibility cleanup ray-traces newer observations to remove penetrated stale cells, addressing lingering wall and overhanging-obstacle artifacts without an additional local implementation.

Prebuilt terrain input

`config/elevation_mapping.yaml` owns mapper inputs, local grid, fusion and cleanup. `config/autonomy_light.yaml` contains the one canonical map/base TF names, calibration and Super-LIO/Livox inputs plus the compatibility-output transport; the launcher binds shared frame names and one publish rate to mapper and bridge. Use `--elevation-config FILE` to select another mapper profile; the output topic is a launcher remap (`AUTONOMY_LIGHT_ELEVATION_MAP_TOPIC`).

`underlying_map_topic` accepts a prebuilt `GridMap` carrying an `elevation` layer. A Super-LIO PCD remains the map used by `--map` relocation and is not silently reinterpreted as terrain. This keeps localization and terrain-map contracts explicit.

Operational commands

- `./build.sh` installs/builds Livox, Super-LIO, Grid Map, vendored KINDR/KINDR ROS, the mapper, and bringup.
- `./launch.sh --vis` opens a latest-frame GridMap elevation viewer; `--rviz` opens the GridMap RViz display.
- `./mapping.sh --output maps/site.pcd` runs Super-LIO full SLAM and writes the optimized PCD at shutdown.